

Fault Tree Analysis for the Communication of a Fleet Formation Flight of UAVs

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Abstract—Fleet flight of drones has been an interesting topic in the last years due to the advantages of the cooperative fleet of UAVs in both civilians and military applications. Fault Tree Analysis (FTA) is a well-known technique used to identify the causes that lead to the failure of a top event. In this paper, we aim to analyze the reliability of communications between drones or between drones and ground station control using FTA. Improving the reliability of the communication system would guarantee the success of the mission. Data for basic events has been gathered from the Nonelectronic Parts Reliability Data Publication (NPRD). The probability of occurrence of communication failure between the drones is calculated taking into consideration the presence or absence of a crash.

Keywords—drones; communication; formation fleet; reliability; fault tree

I. INTRODUCTION

Drones, also known as Unmanned Aerial Vehicles (UAV) or Remotely Piloted Vehicles (RPV), comprise of an aircraft conducted without a pilot. Cooperative UAV reflects a kind of formation flight that is regrouped to achieve a common goal in terms of target position and accomplish a certain trajectory tracking or a mission [1]. They are broadly used in both military and civilian missions in hostile and hazardous environments without risking human life. For this reason, they are recognized as the vehicles for 3D missions that are ‘dirty, dull or dangerous’ [2]. This paper analyzes the reliability of the communication systems of a cooperative UAV fleet and it is organized as follows: Section 2 presents a review of related works. Section 3 provides the proposed fault tree analysis for each communication failure including the description of each intermediate event. In section 4, an analysis approach of the fault tree is shown taking in consideration the probabilities of related basic events. Section 5 presents the simulation results that exhibit the variation of the occurrence probability of the communication failure. A conclusion discussing the results is presented in Section 6.

II. RELATED WORK

Faulty behaviors of the drones or the fleet could be caused by failures, breakdown or malfunction in the components, the computers, and the platform; or in the

information flow between the aerial vehicles. A single failure does not necessarily provoke the failure of the complete system [3]. Equipping each UAV with hardware redundancy would decrease the probability of failure of the vehicle. However, not all UAV could be equipped with redundancy hardware due to limited size, weight and power as well as increased costs [4]. Therefore, fault avoidance is a good solution.

Concerning the communication system, component-level faults comprise those of the sensors, actuators, control surfaces, flight computers, engine and GPS data [5]. Analyzing the failure of UAV has been covered by different methods. In [6], the probability of catastrophic failure has been estimated using a histogram of pre-fault control command distribution. In [7], neural network is used to adapt with the effect of the inversion error. Sliding mode is adopted for a non-linear control permitting it to achieve the mission in a finite time [8]. Founded on the Binary Decision Diagram (BDD), the reliability approach was also used for the mission planning of UAVs [9]. Considered as a slow process, the BDD was improved with the help of an empirical approach, [10]. To isolate the anomaly within a formation flying aerial vehicle, a data-driven approach with a sequence of input and output data pairs was used to detect the failure by spotting an abnormal change [11]. The concept of the model-driven approach, developed by Beard [12], is widely used for fault detection and isolation (FDI). A control system adopting the leader-follower strategy with the problem of the collision avoidance is presented in [13]. In [14], the authors present the different methods of reliability (FMECA, FTA, and FRACAS) used for the Small UAVs (SUAVs). The diagraph analysis reflects another approach for determining the highest cause of occurrence of failure using the graph theory with pseudo Boolean expressions and leading to the shortest path trajectory [15]. In addition to FTA, the Markov analysis (MA) and the Dependence Diagram Analysis (DDA) constitute well-known methods for reliable analysis of aerial systems [16].

III. PROPOSED APPROACH

In this section, a failure analysis of fleet formation flight of UAVs is discussed. Reliability analysis tools such as failure modes and effects analysis (FMEA) and FTA are evident for the reliability of redundancy hardware. This analysis can be adopted in two ways: inductive and

deductive. In the inductive form, the component failures are designed at the lowest level and their effects on the higher level; which is contrary to the deductive form where the causes of each failure are determined.

Several factors cause the failure of a cooperative formation flight including the environmental effects, the damage of at least one of the team, the information flow faults, the obstacles and the collisions of the UAVs in addition to their anomalies. Fig. 1 describes the four intermediate events that influence communication within the fleet. Each event illustrated by a transfer gate, will be described in detail in the following sections.

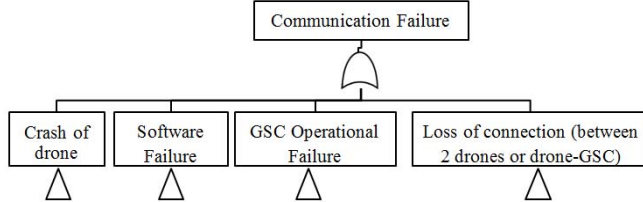


Figure 1. Fault Tree Analysis of the communication failure of fleet of drones

A. Crash of Drone

An environmental event can attribute to the deterioration of the UAV. It can occur to the platform itself, the components or the systems. The crash of the vehicle could be caused due to certain events such as the fleet management system failure, the attack of the drone by a sniper or an electromagnetic pulse, a collision event or even due to a mechanical failure (Fig. 2). Drones should know other drones' positions in order to ensure the fleet management system and avoid the collisions between them. The fleet management system ensures coordination between devices. The inertial measurement unit (IMU) and the inertial navigation system (INS) are used, in addition to GPS and the altimeter, to determine the orientation, position and altitude of the drone. When an UAV submits faulty data and/or moves away from its desired trajectory, it increases the probability of collision with its neighbors and with obstacles

in the environment. This failure could be presented when the proximity sensor is broken or when it gives an inaccurate decision.

Mechanical failure is one of the major causes of a crash (Fig. 3). Mechanical failure comprises failure of either one of the following components: the engine, more than one propeller or the power. Bad and faulty maintenance could also contribute to mechanical failure. Battery failure is attributed due to an overcurrent / undercurrent, physical damage, overheating or exhaustion. On the other hand, the engine can be interrupted due to hardware failure as the servomotor, the actuator and disruption of cables or due to the loss of onboard computers. The cooling system represents an essential factor to avert the engine from dropping out. A catastrophic problem occurs in the event of a crash of more than one propeller.

B. Software and Ground Station Control Operational Failure

In addition to equipment failure, a software failure can occur leading to the disruption of the communication between drones. The reliability of the software is ensured through good design, regular updates of both the operating system and related applications ensuring it is free of viruses/malwares. The main problem occurs when the operation system stops suddenly (Fig. 4).

Failure of the communication Drone-GSC might be due to the transmitter/receiver faults, the operator's behavior and misjudgment of the weather (Fig. 5).

C. Information Flow Faults

The formation of a cooperative fleet is based on inter-vehicle communication based wireless technologies such as Bluetooth, Zigbee, or WiFi with an adhoc network. Each UAV represents a node capable of transmitting and receiving the state (position, velocity, altitude) to and from each one of its neighbors. Each one of the fleet collects the data from its sensors and transmits assigned and visited targets as well as its health status to its teammates.

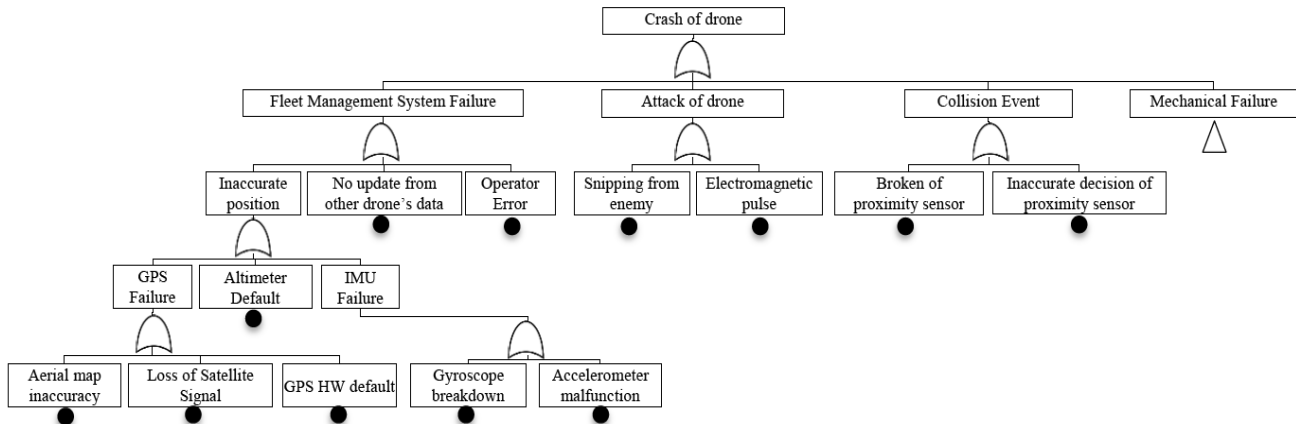


Figure 2. FTA of a crash of drone

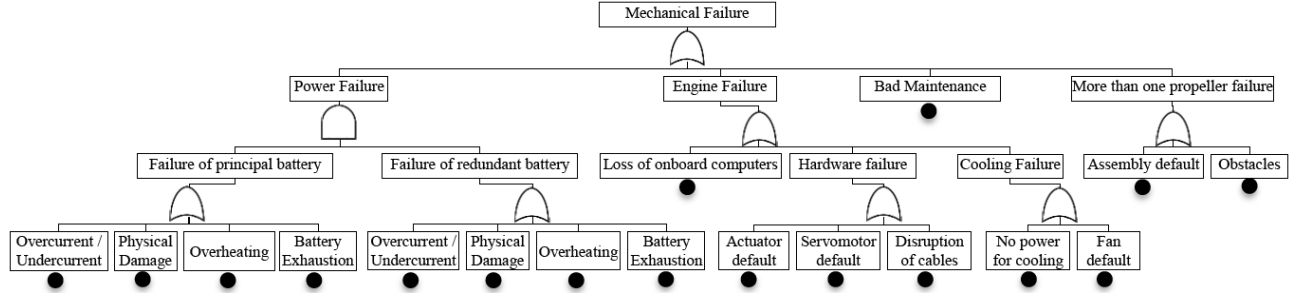


Figure 3. FTA of a mechanical failure of drone

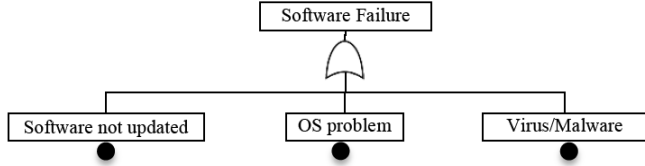


Figure 4. FTA of the software failure of UA

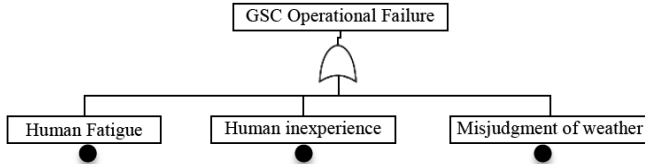


Figure 5. FTA of the software failure of UA

If the medium of communication is exposed to jamming, echoes and noise, then that might interfere with what is transmitted, affecting the overall communication. Other factors that might influence communication are the mobility of UAVs and the presence of nearby users. Information flow is necessary to achieve the mission of the fleet. Faults or loss of information flow are pretended due to nodes loss, damage of an agent, failure of the flight system and the presence of obstacles. It might occur between two or more UAVs or

between the aerial vehicle and the ground control system. The ground control station (GCS) forms a relay point for the information flow, so it can represent a major issue within the overall communication system. Antennas' faults could be due to manufacturing defects or electrical faults. Synchronization of the message is an essential mechanism in the transmission of the message. The distance between the source and the destination can influence the quality of the received information as when the distance increases, the transmitted signal can be exposed the attenuation and atmospheric noises (Fig. 6).

Environmental Effects

The environmental effects include each exterior factor that affects the communication of the fleet. It encloses the natural phenomenon, animals, human, obstacles, and weather. Weather that could have a major influence includes temperature, wind direction, speed, turbulence, clouds, fog, thunderstorms, atmospheric pressure and icing. Icing can increase the weight of the UAV, as it conglomerates affecting the lift and thrust forces. Air density influences the phases of takeoff and climbing. The other factors affect the visibility of the UAV.

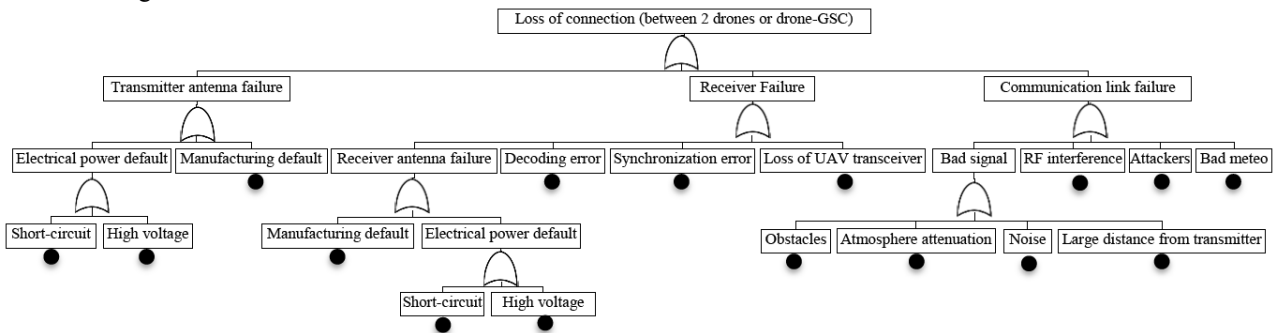


Figure 6. FTA of the loss of connection between Drone - Drone or Drone - GSC

IV. ANALYSIS APPROACH

This section aims to present the analysis probabilistic approach for the top event taking into consideration basic events. A FTA is used in order to estimate the probability of failure of the top event "communication failure" at a certain time. To reach the goal, some reference data can be useful. However, it is not easy to find failure rates for all basic

events. In our work, the Nonelectronic Parts Reliability Data Publication (NPRD-2016) database is used to get data for basic events. For each one, it includes the following reliability information: $\lambda_{\min}$ = minimum failure rate, $\lambda_{\max}$ = maximum failure rate, λ_{mean} = mean failure rate, $t_{\min}$ = minimum operation hours, $t_{\max}$ = maximum operation hours, $N_{\min}$ = minimum number failed and $N_{\max}$ = maximum number failed. Table I extends the failure rates of these events.

TABLE I. FAILURE RATE OF BASIC EVENTS FROM NPRD-2016

Basic Events	Minimum Failure Rate (λ_{min}) ($\times 10^{-6}$)	Maximum Failure rate (λ_{max}) ($\times 10^{-6}$)	Mean Failure Rate (λ_{mean}) ($\times 10^{-6}$)
Operator Error	1.771E+01	6.733E+01	3.191E+01
Broken of proximity sensor	4.290E+00	4.418E+00	3.717E+01
Inaccurate decision of proximity sensor	3.783E-02	4.087E-01	2.891E-01
Altimeter default	2.725E+00	2.067E+01	1.183E+01
Loss of onboard computers	2.786E+00	2.783E+01	2.891E+00
Assembly default	8.992E+01	6.296E+02	2.070E+02
Aerial map inaccuracy	1.007E+00	2.891E+00	1.567E+00
Loss of satellite signal	3.483E-01	4.087E+00	2.891E+00
GPS Hardware Fault	6.306E-01	6.385E+00	1.452E+00
Gyroscope breakdown	6.484E+00	1.794E+01	1.243E+01
Accelerometer malfunction	9.977E-01	7.883E+00	3.504E+00
Overcurrent/Undercurrent	1.669E+01	7.809E+01	3.711E+01
Battery physical damage	3.483E-01	4.087E+00	3.134E+00
Overheating of the battery	1.262E+01	7.782E+02	4.442E+01
Battery exhaustion	3.483E-01	4.087E+00	2.891E+00
Actuator default	1.218E-01	8.569E-01	3.444E-01
Servomotor default	3.575E-01	3.952E+00	2.961E+00
Disruption of cables	4.000E-05	4.087E+00	4.800E-05
No power for cooling	3.483E-01	4.087E+00	2.891E+00
Fan Default	3.483E-01	4.087E+00	2.891E+00
Manufacturing default	3.662E+00	2.047E+01	1.189E+01
Synchronization error	1.317E+00	4.613E+00	2.487E+00
Loss of UAV transceiver	7.285E-01	3.740E+00	9.635E-01
RF interference	4.905E+01	2.491E+02	1.526E+02
Short-circuit	2.352E+00	9.104E+00	4.813E+00
High voltage	1.445E+00	2.044E+00	1.545E+00
Noise	1.741E-01	1.995E+00	1.445E+00

For the other basic events that are not included in the database, such as bad meteo, obstacles, etc. a minimum, maximum and mean probability is considered as shown in Table II.

TABLE II. PROBABILITIES OF BASIC EVENTS

Basic Events	P _{min}	P _{max}	P _{mean}
No update from other drone's data	5.5E-02	3.7E-01	1.4E-01
Snipping from enemy	6.0E-03	1.8E-02	8.0E-03
Electromagnetic pulse	4.6E-03	2.6E-02	1.8E-02
Bad maintenance	4.0E-03	2.2E-02	1.2E-02
Software not updated	4.0E-03	2.0E-02	6.0E-03
OS problem	2.0E-03	8.0E-03	3.0E-03
Virus/Malware	6.0E-03	1.4E-02	1.0E-02
Human Fatigue	1.0E-02	4.0E-02	2.0E-02
Human Inexperience	1.6E-02	6.0E-02	2.4E-02
Misjudgment of weather	1.1E-02	3.6E-02	2.2E-02
Obstacles	1.5E-01	4.0E-01	3.5E-01
Decoding error	1.2E-03	6.8E-03	4.6E-03
Attackers	2.4E-02	8.0E-02	3.0E-02
Bad weather	1.4E-02	6.8E-02	4.7E-02
Atmospheric attenuation	8.0E-03	2.6E-02	1.2E-02
Large distance from transmitter	2.9E-03	1.1E-01	4.3E-03

A Weibull distribution on a total duration of 100 000 hours is considered since it represents the most probabilistic approach used to describe a random distribution of a component lifetime [17]. Our aim is to ensure a successful communication between the UAVs on one side and between the drones and GCC on the other side considering the real conditions of environment, such as large obstacles, bad weather, snipping from enemy, etc. The calculation is based on taking randomly a failure rate λ between λ_{min} and λ_{max} , then using a beta distribution in order to obtain the probability that the event occurs before a time t . The coefficient of dispersion is considered between 0.05 and 0.1 ($0.05 < v_\lambda < 0.1$) for the events of the NPRD database that

depend on a certain timescale. However, all other events are independent as they represent the possibilities of occurring during the mission. Therefore, different coefficients of dispersion, between 0.4 and 0.8, have been considered. In addition, based on Weibull distribution, the probability takes a shape factor β that should be between 1.5 and 4 whose dispersion is realistic ($\beta=3$ in our case). The probability distribution takes two parameters into consideration following these formulas:

$$p = -m_y + \left(\frac{1-m_y}{v_y} \right) \quad (1)$$

$$q = \left(\frac{1-m_y}{m_y} \right) \cdot p \quad (2)$$

where m_y and v_y represent respectively the mean and the standard deviation of a random variable Y .

$$m_y = \frac{\lambda_{mean} - \lambda_{min}}{\lambda_{max} - \lambda_{min}} \quad (3)$$

$$v_y = \frac{\lambda_{max} - \lambda_{min}}{\lambda_{mean} - \lambda_{min}} \quad (4)$$

The following equation represents the Weibull distribution:

$$P(Y < y) = 1 - e^{-\left(\frac{y}{\eta}\right)^\beta} \quad (5)$$

V. COMMUNICATION FAILURE PROBABILITY

Using the Weibull distribution, the probability of having communication failure between the UAVs is calculated. Two cases are considered: (1) communication failure that is depends on the four intermediate events of Fig. 1 and (2), being the case that excludes the crash of drone event. In case 1 (Fig. 7), it can be seen that the communication failure depends on time and takes the shape of the crash of drone as most of the events are those of the components failure. The communication is successful with some minor defaults until it reaches 40 000 hours. The probability of occurrence of the failure increases till it attains it maximum at 63 000 hours. From this time, the exchange of information between the drones or between the drone and the ground station will be stopped.

In case 2; we avoided mechanical failures, the collision, the fleet management system failure and the attack. The communication is influenced by the loss of connection where the events are independent on time (Fig. 8). The probability of occurrence of the communication failure has a maximum of 0.2 (Fig. 9).

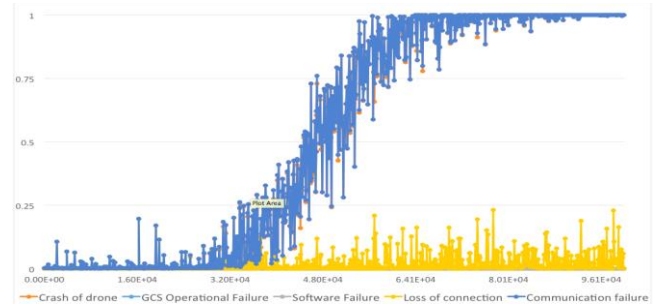


Figure 7. Probability of communication failure between UAVs depending on crash of drone

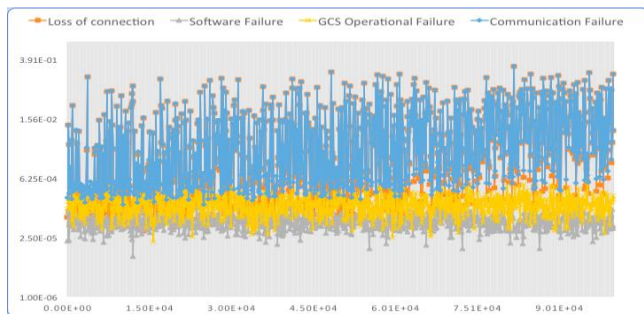


Figure 8. Probability of communication failure between drones without having a crash

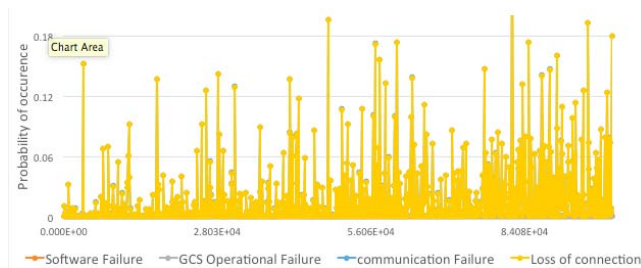


Figure 9. Probability of occurrence of a communication failure

VI. CONCLUSION

Several fault tree analysis are considered showing the causes of communication failures. The crash of a drone is the major cause of the communication failure. Moreover, loss of communication can also occur due the loss of connection between the transmitter and the receiver with a probability of occurrence of 0.2. Further studies should be conducted to find suitable solutions that should decrease this probability and ensure more reliable communication.

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